

Abstract Algebra

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1 Tues January 13

One midterm one final. Midterm on groups, final mostly on rings.

1.1 Introduction

Motivating question: consider a triangle. How many symmetries are there?

The answer turns out to be 6 through rotations and reflections, just the permutations of the vertices. We call this set D_6 , with composition as the operation. Together, this makes a **group**. Note that all elements of D_6 can be written as compositions of rotations and reflections. We can then say that rotation and reflection **generate** the group D_6 .

Definition 1.1

A **group** is a set G equipped with a binary operation $\cdot : G \times G \rightarrow G$ satisfying the following axioms:

1. Associativity: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
2. Identity: $\exists 1 \in G$ s.t. $a \cdot 1 = a$.
3. Inverse: $\forall a \in G, \exists a^{-1}$ s.t. $a^{-1} \cdot a = 1$
4. Closed: $\forall a, b \in G, a \cdot b \in G$. (This is kind of implicit in our definition of $\cdot$).

Some other groups/rings:

- $(\mathbb{Z}, +)$
- $(\mathbb{Q} \setminus 0, \times)$
- $(\mathbb{Z}, +, \times)$: Ring
- $(\mathbb{Z}/12, +)$
- S_3 : all the permutations of 3 elements

2 Thu Jan 15

Today we will prove some basic properties of groups.

Proposition 2.1

Let $(G, \cdot)$ be a group.

1. The identity is unique.
2. For any $g \in G$, g^{-1} is unique.
3. For any $g \in G$, $(g^{-1})^{-1} = g$.
4. $(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$.
5. Associativity works for any n

Proof. Just do the obvious.

Definition 2.1

The **order** of an element $g \in G$ is the smallest $n \in \mathbb{Z}_{>0}$ such that $g^n = e$.
The **order** of a group G is its cardinality $|G|$.

Definition 2.2

A subset $S \subseteq G$ **generates** the group G if

$$\{\text{iterative products of elements of } S \text{ and their inverses}\} = G.$$

For example, $\{1\}$ generates $\mathbb{Z}$. A word is just an expression like r^2s . A relation in G is a word whose underlying element is the identity. We write $G = \langle S \mid R \rangle$ if G is generated by S and has relations R .

Definition 2.3

The dihedral group D_{2n} is the set of symmetries of a regular n -gon.

$$D_{2n} \text{ has presentation } \langle r, s \mid r^n = e, s^2 = e, rs = sr^{-1} \rangle.$$

Definition 2.4

A group is **Abelian** if $\forall x, y \in G$,

3 Tue Jan 20

You know what a bijection is. Let Ω be a set. $\text{Bij}(\Omega)$ with composition is a group. This is really the group of permutations on Ω . This group is called S_n , the symmetric group on n elements. Order of $\sigma \in S_n$ is the least common multiple of the lengths of its disjoint cycles.

S_n is an example of a **Weyl group**.

Definition 3.1

A **field** is an abelian group with identity for $+$ 0 such that $F \setminus \{0\}$ is an abelian group under a different operation $\cdot$ such that $a \cdot (b + c) = a \cdot b + a \cdot c$ for any $a, b, c \in F$.

Definition 3.2

Let $GL_n(F)$ be the group of invertible $n \times n$ matrices with entries in F .

4 Thu Jan 29

Definition 4.1

A group is **cyclic** if it is generated by a single element.

Cyclic groups are Abelian.

5 Thu Feb 5

Theorem 5.1 First Isomorphism Theorem

Let G be a group and $\varphi : G \rightarrow H$ be a homomorphism. Then $\ker \varphi \leq G$ is a normal subgroup and $G / \ker \varphi \cong \varphi(G) := \text{Im } \varphi$.

In general, if we suspect $G/K \cong H$, then we try to write a map $G \rightarrow H$ with kernel K .

Definition 5.1

Let A_n (alternating group on n letters) be the kernel of $\text{sgn} : S_n \rightarrow (\{\pm 1\}, \times)$.

Then $S_n/A_n \cong \mathbb{Z}_2$. We want to find normal subgroups $K \triangleleft G$ and study K and G/K in place of G .

Definition 5.2

A group G is **simple** when its only normal subgroups are $\{e\}$ and G itself.

Then, for example, A_5 and $\mathbb{Z}_p$ are simple groups.

6 Tue Feb 10

Theorem 6.1 Second Isomorphism Theorem

Let G be a group with subgroups A and B . Suppose $A \leq N_G(B)$. Then $AB \leq G$, and $B \trianglelefteq AB$, and $A \cap B \trianglelefteq A$, and we have an isomorphism $AB/B \simeq A/A \cap B$.

Proof. $AB \leq N_G(B)$ implies AB is a group, $B \trianglelefteq AB$, and $A \cap B \trianglelefteq A$. This is because we can write $abab = a(ab)b$. Also, B is normalized by B and by A . So B is normalized by AB . Also if $x \in A \cap B$, then $a^{-1}xa \in B$ since $x \in B$, $a \in N_G(B)$. And we also automatically have $a^{-1}xa \in A$ so $a^{-1}xa \in A \cap B$.

Now we construct the composition $\varphi : A \rightarrow AB/B$ as $A \rightarrow AB \rightarrow AB/B$. By the first isomorphism theorem, we have $A/\ker \varphi \simeq AB/B$. Suppose $\varphi(a) = e$ for $a \in A$. We wish to show that $a \in B$. This means $aB = B$ as cosets $\iff a \in B$. Thus $A/A \cap B \simeq AB/B$.

Theorem 6.2 Third Isomorphism Theorem

Let G be a group, and H and K normal subgroups of H with $H \leq K$. Then $K/H \trianglelefteq G/H$ and $(G/H)/(K/H) \simeq G/K$.

Proof. Convince yourself that $K/H \trianglelefteq G/H$. We define $\varphi : G/H \rightarrow G/K$ by $gH \mapsto gK$. Then $\ker \varphi = \{gH \in G/H \mid \varphi(gH) = 1K\} = \{gH \in G/H \mid gK = K\} = \{gH \in G/H \mid g \in K\} \simeq K/H$.

Theorem 6.3

Let G be a group and $N \trianglelefteq G$. There's a bijection from subgroups $A \leq G$ containing N to $\bar{A} = A/N$ of subgroups of G/N . Furthermore, for any A, B on the LHS:

- $A \leq B \iff \bar{A} \leq \bar{B}$
- If $A \leq B$, $[B : A] = [\bar{B} : \bar{A}]$
- $\langle \overline{A, B} \rangle = \langle \bar{A}, \bar{B} \rangle$
- $\overline{A \cap B} = \bar{A} \cap \bar{B}$
- $A \trianglelefteq B \iff \bar{A} \trianglelefteq G/N$

Recall $[B : A]$ is the number of A -cosets in B .

Proof. Let $\varphi : G \rightarrow G/N$. Given any subgroup $\bar{A} \leq G/N$, $\varphi^{-1}(\bar{A})$ is a subgroup. Given any subgroup $A \leq G$ containing N , $\varphi(A) \leq G/N$. Note that $\bar{A}_1 \leq \bar{A}_2 \iff \varphi^{-1}(\bar{A}_1) \leq \varphi^{-1}(\bar{A}_2)$.

7 Thu Feb 12

Proposition 7.1

Suppose G is a finite abelian group and p is a prime with $p \mid |G|$. Then G has an element of order p .

Theorem 7.1 Jordan-Holder

Every finite group admits a composition series, and there is a permutation $\pi \in S_r$ such that the quotients are the same for any composition series up to isomorphism.

Theorem 7.2

If $H \leq G$ with $[G : H] = 2$, H is normal.

Theorem 7.3

Suppose p is the smallest prime dividing $|G|$. Then any index p subgroup is normal. That is, $[G : H] = p \implies H \trianglelefteq G$.

8 Thu Feb 26

Review for midterm. Our topics are:

- Groups, Subgroups, Lagrange's Theorem
- Homomorphisms and Quotients
- Actions and the Class Equation
- Classification (Jordan-Holder, Automorphisms, Sylow)

Definitions we should know:

- Definition of a group, subgroup, quotient group, group action
- Cyclic, normal, homomorphism, isomorphism, images and kernels,
- Coset, orbit/stabilizer, faithful, transitive, conjugacy class, p-group, Sylow p-subgroup

Definition 8.1

An action G on X is **faithful** if the corresponding map $G \rightarrow S_X$ is injective.

In other words, if some group element does nothing, that group element must be the identity.

Definition 8.2

An action is transitive if $\forall x, y \in X, \exists g$ such that $g \cdot x = y \iff X$ is a single G -orbit.

If an action is faithful and transitive, G should look like X .

Centralizers are a special case of stabilizers

$H \leq G$ iff $\forall h_1, h_2 \in H, h_1 h_2^{-1} \in H$.

If you have two subgroups and one normalizes the other, their product is G .

$\langle a^{n/d} \rangle$ is a subgroup of order d in $\mathbb{Z}_n$.

Example 8.1

Compute $|C_{S_4}(123)|$.